

Federal Education Spending and State-Level Investment Patterns (2000 - 2022)



Content *Overview*

I Motivation →

II Data Sourcing & EDA →

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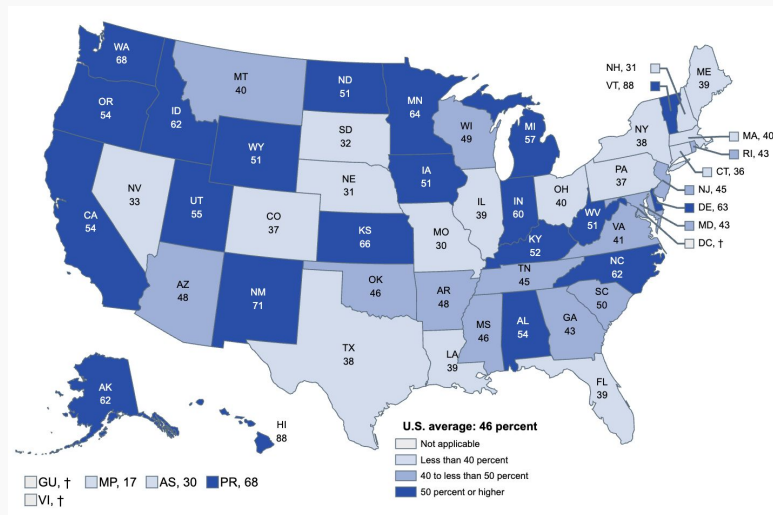
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How does federal funding share correspond to lower or higher total per-student spending?

1. The federal government provides ~8–10% of K–12 education funding nationally (U.S. Department of Education, 2023).
2. States differ dramatically in their capacity to fund education, creating large disparities in per-student resources (NCES, 2023).
3. States with higher federal share often have lower tax capacity and higher concentrations of poverty, shaping total spending levels (Brookings Institution, 2020).
4. Understanding how federal funding relates to total spending helps explain equity gaps across states (Education Commission of the States, 2022).



State Revenue for Public K – 6 Schools, National Center for Education Statistics, 2024.

II. Data & Processing

Dataset:

- **Source:** NCES Annual State Education Finance Survey (2000–2022)
- **Units of analysis:** State-year observations (~1,200 rows)
- Key variables extracted:
 - Federal revenue
 - total revenue
 - Enrollment
 - Year
 - State name/fips

Cleaning & Standardization

Remove
commas/sy
mbols from
revenue/enr
ollment

Convert
financial
variables to
numeric

Dropped
missing
values or
invalid totals

Standardize
state names

Constructed Variables

Federal per Capita =
federal revenue / enrollment

Total per Capita =
total revenue / enrollment

Federal Share (%)=
federal revenue / total revenue

Scaled per-capita measures
into thousands

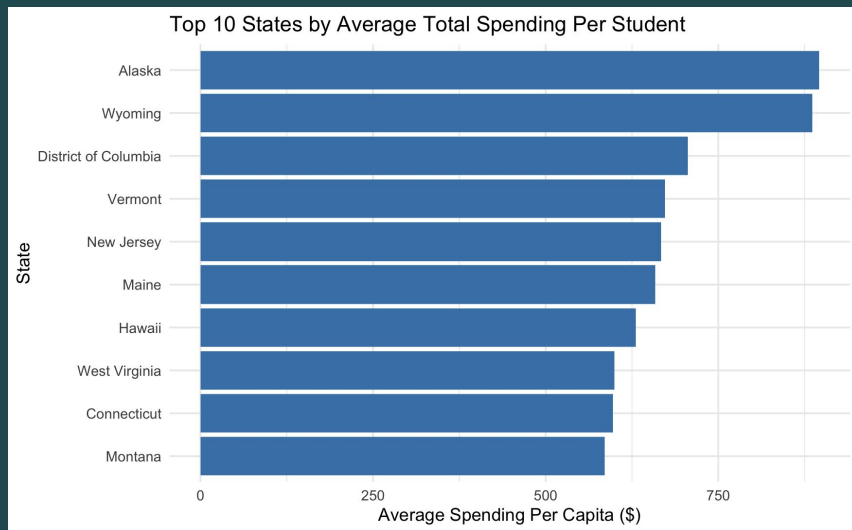
Final Analysis Dataset

Includes:

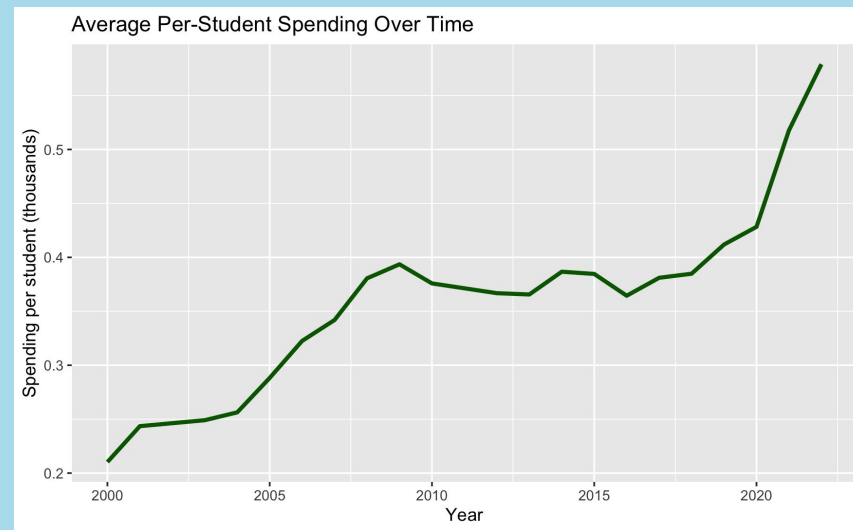
- State
- Year
- Total per-capita (k)
- Federal per-capita (k)
- Federal share

Provides for consistent, comparable financial indicators across 22 years

II. Exploratory Data Analysis



- States with the highest per-student spending are small, rural, or high-cost states
- Top-spending states cluster well above the national average, showing that state wealth, local revenue capacity, and cost-of-living differences drive large disparities in education funding.



- Per-student spending generally increases from 2000 through 2022, reflecting rising education costs, inflation adjustments, and expanded service needs over time.
- Spending levels flatten between 2009–2015, aligning with recession-era budget constraints and slow fiscal recovery across many states

III. Frequentist Regression: Does Federal Share Predict Spending?

Key Findings

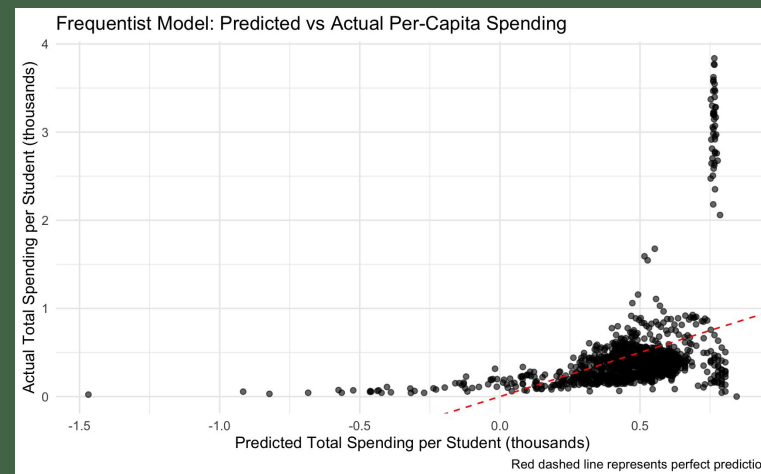
- Higher federal share predicts *lower* total spending, even after controlling for year ($\beta = -0.025, p < .001$).
- Year also has a small negative effect, indicating a slight downward trend over time when holding federal share constant.
- The model explains ~15% of variation in spending (meaningful, but shows that states differ in ways the model doesn't capture).

Regression Coefficients

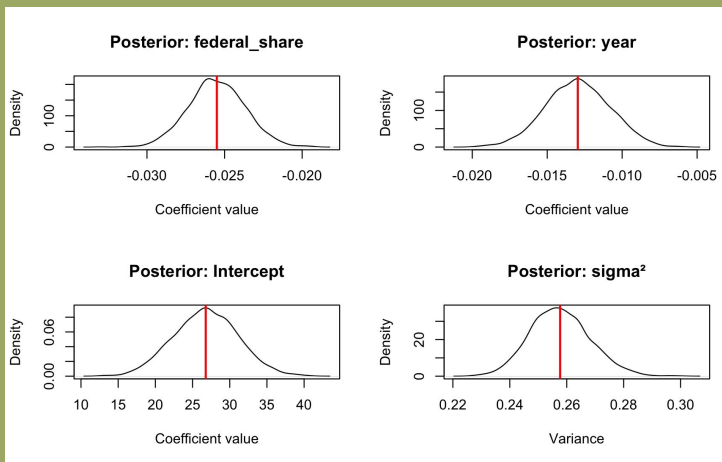
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	26.796930	4.379066	6.119	1.27e-09	***
federal_share	-0.025492	0.001853	-13.759	< 2e-16	***
year	-0.012970	0.002176	-5.959	3.33e-09	***

Model Fit: Predicted vs. Actual Spending



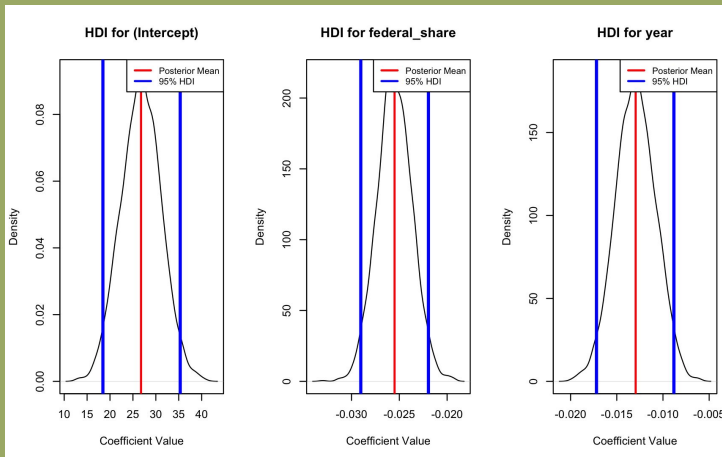
III. Bayesian Regression: What do the Posterior Distributions Show?



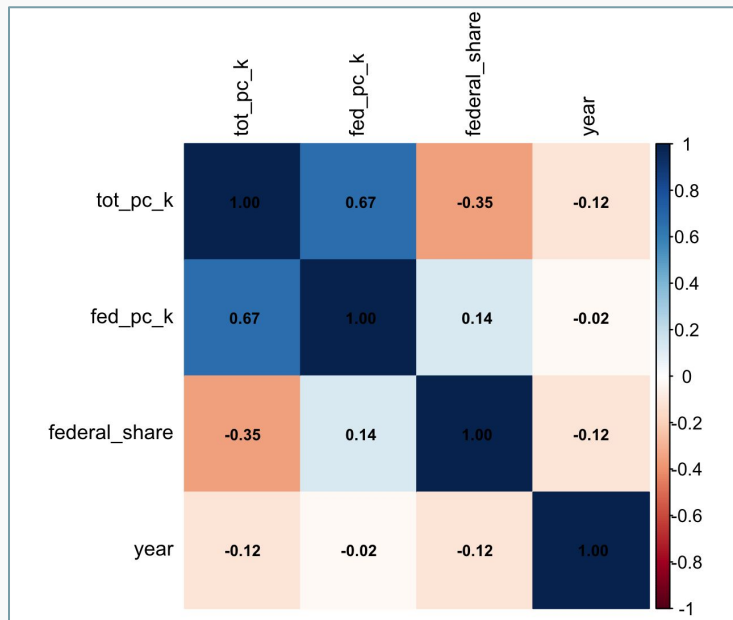
Key Takeaways:

	Mean
(Intercept)	26.77292
federal_share	-0.02550
year	-0.01296
sigma2	0.25768

- Posterior distributions show strong evidence of negative effects.
 - The posterior distributions for federal_share and year are centered *below zero*, indicating a high probability that both predictors *decrease* per-student spending.
- 95% of HDIs do not cross zero.
- Reinforces the main conclusion that states with a higher federal share of revenue tend to have lower total spending per student.



IV. Results & Interpretation



Correlation Structure Among Key Variables

Key Findings from Regression Models

Higher federal share predicts lower total spending ($\beta \approx -0.025$; $p < .001$).

This reflects structural fiscal inequality: states relying most on federal funds tend to have lower overall revenue capacity.

Time trend is slightly negative, but very small ($\beta \approx -0.013$), meaning year effects explain little of the variation.

Model fit is modest ($R^2 \approx 0.15$), indicating spending disparities are driven by deeper structural factors beyond federal share and year

Overall, the results show that federal reliance signals lower state investment rather than federal aid causing lower spending.

V. Limitations /Next Steps

LIMITATION: State-Level Aggregation

State averages hide large district-level inequalities and differences in cost of living, which limits how precisely spending patterns can be interpreted.

01

LIMITATION: Missing Structural Factors

Important drivers of funding (such as poverty rates, tax capacity, demographics) were not included, which narrows the model's explanatory power.

02

NEXT STEP: Build a More Complete Equity Model

- Incorporate district-level finance data, adjust for cost of living, and add socioeconomic variables.
- Use fixed-effects or longitudinal models to better isolate structural disparities.

03

POLICY APPLICATION: Targeting Underfunded States and Districts

- **Highlight states with structural funding gaps.** Results help identify states where low spending reflects limited revenue capacity.
- **Support more equitable funding formulas.** Findings can guide federal and state policymakers in adjusting formulas to better target high-need regions.
- **Inform smarter federal investment.** Analysis shows where additional federal dollars are most likely to make meaningful improvements in student resources.

04

Sources

- **Brookings – How Federal Investments in Education Research Help Students Succeed**
Wong, V. C., Tipton, E., & Spybrook, J. (2025, April 10). *How federal investments in education research help students succeed*. Brookings Institution.
<https://www.brookings.edu/articles/how-federal-investments-in-education-research-help-students-succeed/>
- **NCES – Public School Revenue Sources**
National Center for Education Statistics. (2024). *Public school revenue sources*. U.S. Department of Education, Institute of Education Sciences. Retrieved from <https://nces.ed.gov/programs/coe/indicator/>
- **Education Commission of the States – 50-State Comparison (Archive)**
Fischer, A., Duncombe, C., & Syverson, E. (2021). *50-State comparison: K-12 and special education funding 2021 (archive)*. Education Commission of the States.
<https://www.ecs.org/50-state-comparison-k-12-and-special-education-funding-2021-archive/>
- **U.S. Department of Education – Federal Role in Education**
U.S. Department of Education. (2025, June 17). *Federal role in education*.
<https://www.ed.gov/about/ed-overview/federal-role-in-education>

Thank you.

Content *overview*

1

Add section title

2

Add section title

3

Add section title

4

Add section title

5

Add section title

6

Add section title

Our *team*



TITLE
Full Name

Write a brief bio mentioning
their background and area of
expertise.

name@example.com



TITLE
Full Name

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expertise.

name@example.com



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Our *team*



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Full Name
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name@example.com



Full Name
TITLE
name@example.com



SPOTLIGHT

Full Name

TITLE

Write a *brief bio* mentioning their background and area of expertise.

Contact
name@example.com

OUR MISSION

This is your mission statement, your chance to *articulate your vision*. In a few words, identify what you do, why you do it, and who you do it for. *Be concise. Be confident.*



Add a few words expanding on the above. You can explain why this mission is important to you and your company and how you'll achieve it.

“Add a quote that captures your *mission statement*. Announce the impact you want to achieve and who you aim to help. Show where your *company is heading*, the ultimate goal.”



Full Name
Title

“Add a quote that captures your *mission statement*. Announce the impact you want to achieve and who you aim to help. Show *where your company is heading*, the ultimate goal.”



Full Name
Title



THE PROBLEM

What pain point are you tackling?
What gap are you going to fill?

Briefly describe the impact of the problem and why people should *care*.

THE SOLUTION

Explain what your company or product does, and why it's critical in the current market.

01

VALUE PROPOSITION

Describe a key benefit that your company, product, or service offers to customers.

02

VALUE PROPOSITION

Define what distinguishes your company, product, or service from what's available.

03

VALUE PROPOSITION

Identify your customers and how you'll connect to their behaviors and needs.

04

VALUE PROPOSITION

Copy this slide and add as many items as necessary to fully outline your company's value.

VALUE PROPOSITION

Identify your company's primary benefit to customers and what makes your business or product distinctive.

Company Name

AUDIENCE

Write a *keen insight* about your persona or target audience.

Needs and motivations

What does your persona want to achieve? What are their goals?

Pain points

Identify an obstacle your persona struggles with. What interferes with their needs and motivations?

How we can help

Explain where your product or service fits.



Full Name

Introduce your audience persona. Highlight some key traits and mention their age, profession, and location.



Full Name

Age	00
Occupation	Job title
Status	Marital status
Education	Highest degree
Location	Place name
Archetype	Book club

“Quote the persona on their needs and wants. This can be a fictional line or an excerpt from a real user interview.”

Bio

Introduce your persona, including who they are and where they come from. Think about a hypothetical user or customer who represents an audience segment.

Personality

Extrovert	Introvert
Spontaneous	Planner
Collaborative	Independent

Current emotions

Anxious	Optimistic
Nervous	Enthusiastic

Goals

- What do they want to achieve and why?
- What do they want to achieve and why?
- What do they want to achieve and why?

Frustrations

- What is keeping them from their goals?
- What is keeping them from their goals?
- What is keeping them from their goals?

Favorite brands

LOGO	LOGO	LOGO
------	------	------

COMPETITIVE LANDSCAPE

Write a *high-level insight* about the competitive ecosystem

	COMPETITOR 1	COMPETITOR 2	COMPETITOR 3	COMPETITOR 4
FEATURES	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more
STRENGTHS	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more
WEAKNESSES	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more

THE OPPORTUNITY

Based on your *competitive analysis* in previous slides, describe the *growth opportunities for your business*. Write a comprehensive claim here, then break it down into individual learnings.

Key learning 1

Go deeper into a specific insight from the competitive analysis and what it means for your business.

Key learning 2

Point to the gap your product will fill and explain how your business can claim that space.

Key learning 3

Consider your brand and customer segments. Explain why you're ideally positioned to reach them.

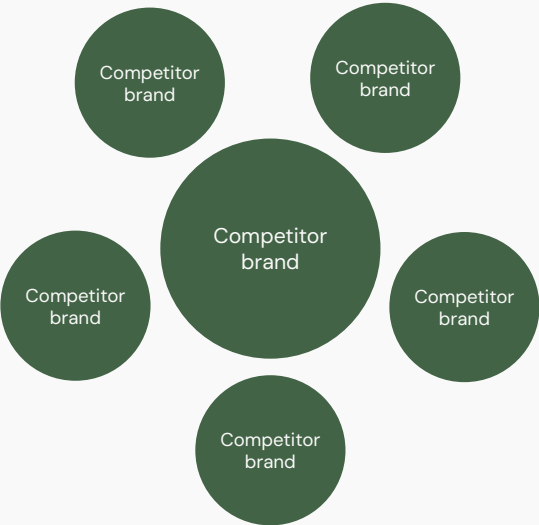
Market analysis

Context

Provide some insight into the state of the market today. Describe any challenges, forces, or factors influencing the market. Share your company's perspective on where the market has been and where it's headed.

Market value	00
Historic growth rate	00
Projected growth rate	00
Cost of entry	00

Key players



Customer segments

Segment 1	Segment 2
Segment 3	Segment 4

Pricing strategy

PRICE POINT			
MARKET	High	Medium	Low
	Premium		
	Mid-level		
	Economy		

BUSINESS MODEL

Clearly outline your business model. This is how you'll create, deliver, and capture value.

Cost structure

What are the major costs of your proposal? Think about salaries, rent, or marketing.

→ Expense 1

→ Expense 2

→ Expense 3

Revenue streams

Outline how you plan to generate revenue through sales, subscriptions, or other methods.

→ Revenue 1

→ Revenue 2

→ Revenue 3

Key activities

What are the core activities your company undertakes to deliver its value proposition? Think about production, research, or fabrication.

Key partnership

Who are your key partners or suppliers? What resources do they provide? And how do they contribute to your business model?

Customer segments

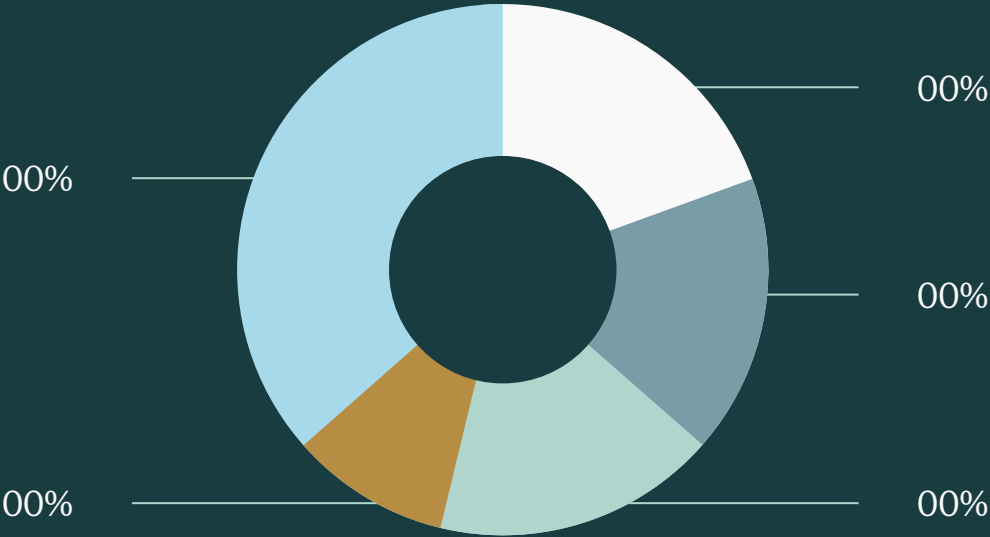
Use this space to restate or prioritize specific customer segments based on your business model.

→ Customer segment

→ Customer segment

→ Customer segment

COST STRUCTURE



- ◆ Staffing
- ◆ Taxes
- ◆ Marketing
- ◆ Logistics
- ◆ Technology

-00%

Estimate potential cost savings in an area or department.

-00%

Add any other potential cost savings in an area or department

If you anticipate changes to your expenses, mention the impact to your cost structure here.

REVENUE MODEL



Revenue stream 1

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 2

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 3

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 4

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 5

Define how you'll generate income from this source. Identify relevant pricing or channels.

Company Name

KEY ACTIVITIES

Primary activity *name*

Describe the core activity of your business, how you create and deliver your products or services. This might be manufacturing, product design, software engineering, customer service, marketing, or project management.

Support activity name

Break down your primary activity into the additional activities required to successfully deliver your value proposition. These should contribute to your core business function.

Support activity name

Think about your everyday operations. For example, if your primary activity is software engineering, your support activities may be customer support, software integration, or maintenance.

Support activity name

Show your audience what a day in the life of your organization looks like. Help them understand what's required to do what you do.



NEXT STEPS

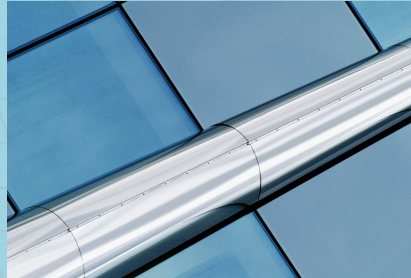
Invite your audience to *learn more* about your organization and to *keep in touch*.

Contact
Full Name
Title
name@example.com

example.com

NEXT STEPS

Invite your audience to *learn more* about your organization and to *keep in touch*.



Contact
Full Name
Title
name@example.com

example.com